

Fig. 7: MCU power management using its non-maskable interrupt

ter its low-power state, where all the peripherals of the MCU, except the RTC, are 'off.'

When system reset is de-asserted, the MCU has to return to its normal state where all its peripherals are active and the CPU can work at maximum computational power. Thus the MCU has to be put in either normal state or low-power state depending on the NMI pin status. Request response will be faster in this case than in the previous command-based approach.

Using software technique. Any of the following two approaches can be used: super loop architecture or the more sophisticated RTOS.

Super loop. The super loop architecture has two main loops—the foreground comprises interrupt service routines (ISRs) that handle asynchronous external events in a timely fashion, and the background is an infinite loop that uses the remaining CPU cycles to perform the less time-critical processing.

To employ LPM, the background loop must first determine that all external and internal events have been processed so that the CPU clock can be stopped until the next external event (an interrupt) wakes up the CPU. This is called an idle condition and is shown in Fig. 8.

Because determination of the idle condition involves testing the variables shared with the foreground (ISRs), the background loop must disable

an ISR could produce new work for the background loop, thus invalidating the idle condition.

By the simplistic nature of the foreground/background processing, the background loop always resumes at the point it was interrupted, so the background loop would enter the LPM while the MCU would have urgent work to do. The MCU is stopped until the next interrupt wakes it up. Thus enabling interrupts before transitioning to a low-power state opens a time window for a race condition between any enabled interrupt and transition to the LPM.

Entering a sleep mode while interrupts are disabled poses a chicken-and-egg problem for waking the system up, because only an interrupt can terminate the LPM. To operate in super loop architecture, the MCU must allow entry into LPM and enabling of the interrupts at the same time without creating this race condition.

Many MCUs indeed allow such an

interrupts before detecting the idle condition. Moreover, the idle condition remains valid only as long as interrupts remain disabled. If the interrupts were enabled after the background loop determines that all its work is done, but before actually switching to the LPM, an interrupt could preempt the background loop at this point and

atomic transition to the sleep mode. Other MCUs support multiple levels of disabling interrupts and can accomplish low-power transitions with interrupts disabled at one level. Yet, other MCUs don't provide any way to enter the low-power mode with interrupts disabled and require different approaches.

RTOS. The super loop software design requires transition to a low-power state to be atomic, or at least interrupt-safe. This requirement does not apply when a more sophisticated architecture, such as a pre-emptive kernel or an RTOS, is used. A pre-emptive kernel executes a special idle task when no other tasks are ready to run because all are blocked waiting for events.

Most kernels provide a way to customise the idle task (using callback functions or macros), so transition to a low-power state inside the idle task can be implemented conveniently. The main difference between a pre-emptive kernel and a foreground/background system is that as long as tasks are ready to run, the kernel doesn't switch the context back to the idle task. Consequently, transition to an LPM is much simpler, because it doesn't need to occur with interrupts disabled.

Unfortunately, a pre-emptive RTOS isn't always an option for a low-end MCU, which simply might not have enough ROM/RAM to accommodate a pre-emptive RTOS.

Low-power features of commercial MCUs

Low-voltage operation, minimised leakage current, sleeping brownout detection (BoD), ultra-low-power 32kHz crystal oscillator, intelligent peripherals, various LPMs and atomic transition to LPMs are key areas of concern for selection of a proper ultra-low-power MCU. These concerns are briefly outlined below with reference to some popular MCUs available in the market.

MSP430. MSP430 provides five power-down modes.

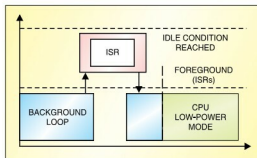


Fig. 8: Super loop system with an LPM